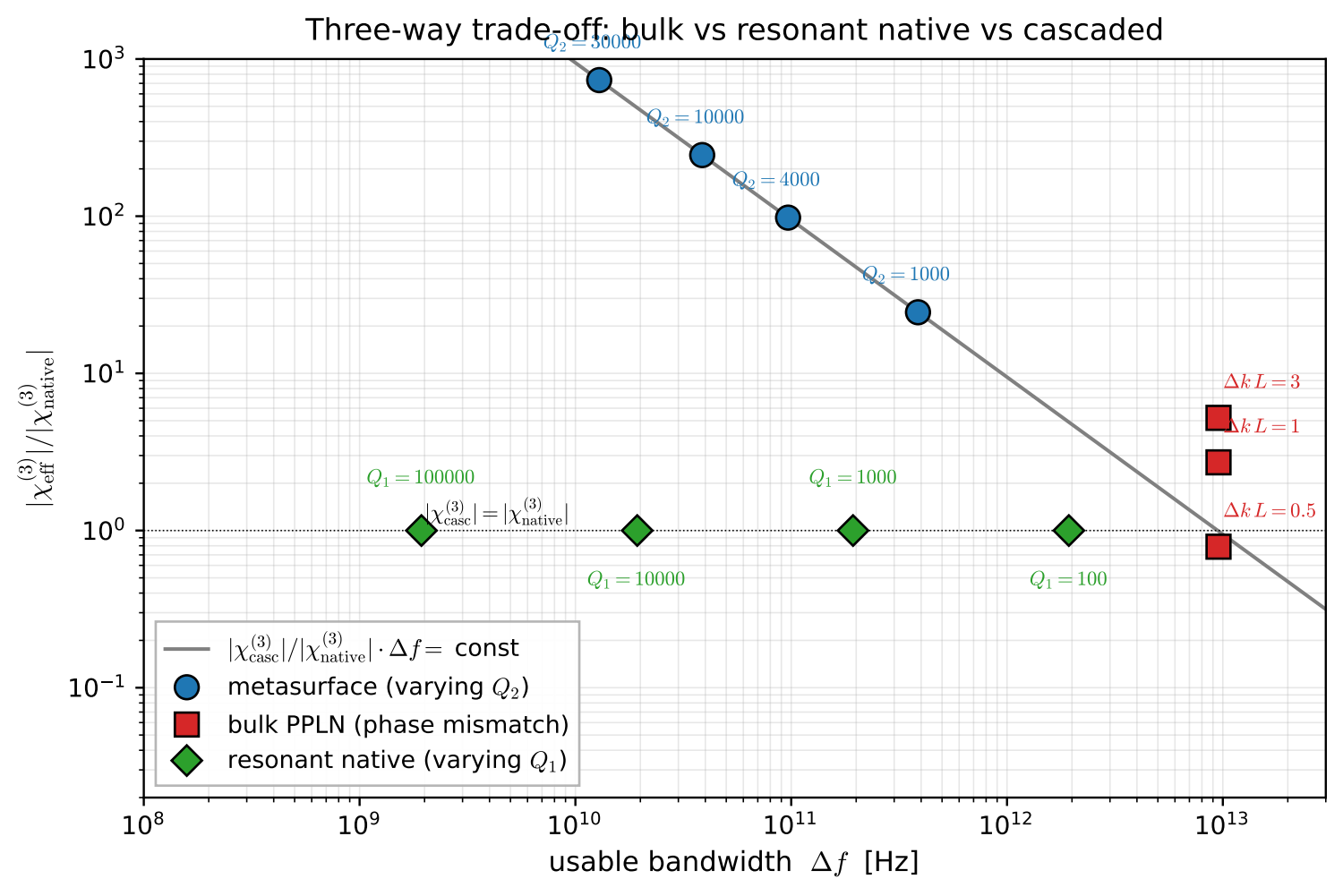
Three-way trade-off: bulk vs resonant native vs cascaded
$Q_2 = 30000$
$Q_2 = 10000$
$Q_2 = 4000$
$Q_2 = 1000$
$\Delta k\,L = 3$
$\Delta k\,L = 1$
$\Delta k\,L = 0.5$
$Q_1 = 100000$
$|\chi^{(3)}_{\mathrm{casc}}| = |\chi^{(3)}_{\mathrm{native}}|$
$Q_1 = 10000$
$Q_1 = 1000$
$Q_1 = 100$
$|\chi^{(3)}_{\mathrm{casc}}|/|\chi^{(3)}_{\mathrm{native}}| \cdot \Delta f =$ const
metasurface (varying Q_2)
bulk PPLN (phase mismatch)
resonant native (varying Q_1)
$|\chi^{(3)}_{\mathrm{eff}}|/|\chi^{(3)}_{\mathrm{native}}|$
usable bandwidth Δf [Hz]
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
10^{8}
10^{9}
10^{10}
10^{11}
10^{12}
10^{13}